FINAL PROJECT REPORT • DATA SCIENCE & MACHINE LEARNING

Rakuten France Multimodal Product Classification

Data exploration, shared preprocessing, text and image fine-tuning, Grad-CAM interpretability, and out-of-fold late fusion on 84,916 e-commerce listings.

Key Results

- **Shared split and out-of-fold fusion:** All unimodal models are trained exclusively on the 67,932 training products and evaluated on the same 16,984 validation products. Fusion weights and the Random Forest meta-classifier are fitted and scored via 5-fold stratified cross-validation on the validation probabilities.
- **Final multimodal score:** Random Forest stacking over CamemBERT, calibrated LinearSVC, and ConvNeXt-Tiny reaches **0.8988 weighted F1** (**0.8856 macro F1**, 1,715 errors), reducing errors by 17.6% vs. CamemBERT (2,082) and 35.4% vs. LinearSVC (2,656).
- **Products without descriptions:** On the 35.0% of validation products lacking a description, text models lose 6.6 to 9.7 F1 points, whereas multimodal Random Forest fusion recovers **+3.98 F1 points over CamemBERT** and **+9.60 points over LinearSVC**.
- **Image model progression:** Moving from a frozen ResNet18 baseline (0.4818 weighted F1) to a full-backbone ConvNeXt-Tiny with smart white-border cropping, layer-wise learning rates, minority-aware sampling, and horizontal-flip TTA reaches **0.6998 weighted F1** (0.6752 macro F1).
- **Interpretability:** LinearSVC weights, ConvNeXt Grad-CAM maps (`model.features[7]`), and Random Forest Gini importances (66% CamemBERT, 23% SVM, 11% ConvNeXt) explain both multimodal gains and remaining failure modes.

Dataset: Rakuten France Multimodal Challenge
Authors: Wahaj Aslam, Bernhard Quehl
Repository: `liora-capstone`

Evaluation: Fixed stratified 80/20 split (16,984 validation products)
Models: Calibrated LinearSVC, CamemBERT, ConvNeXt-Tiny, Random Forest fusion
Date: November 2026

1. Summary and Problem Definition

Online marketplaces receive millions of seller-written product listings that must be placed into the right catalog category for search, filtering, and recommendations. The **Rakuten France Product Classification Challenge** asks models to classify listings into **27 product type codes** (`prdtypecode`) using a French product title (`designation`), an optional product description (`description`), and one product image [notebook `01_dataviz_and_audit.ipynb`].

84,916	27	35.1%	0.8988
LABELLED PRODUCTS	CATEGORIES	MISSING DESCRIPTIONS	FINAL WEIGHTED F1

Text carries most of the predictive signal, but fails when titles are short, generic, or lack condition details. Images are weaker alone, yet provide complementary visual cues—especially on book covers, magazines, and boxed games where descriptions are frequently missing.

As a lower reference point, the **Zero-Rule Majority Class Baseline** predicts the largest category (Pool, code `2583`) for every validation row [notebooks `03_modeling_image.ipynb` , `03_text_baseline.ipynb`]. On the 16,984 validation products, it reaches **12.02% accuracy**, **0.0258 weighted F1**, and **0.0080 macro F1** (14,942 errors). Combining calibrated LinearSVC, CamemBERT, and ConvNeXt-Tiny via a Random Forest reaches **0.8988 weighted F1** and **0.8856 macro F1** (1,715 errors) [notebook `04_text_and_image_fusion.ipynb`].

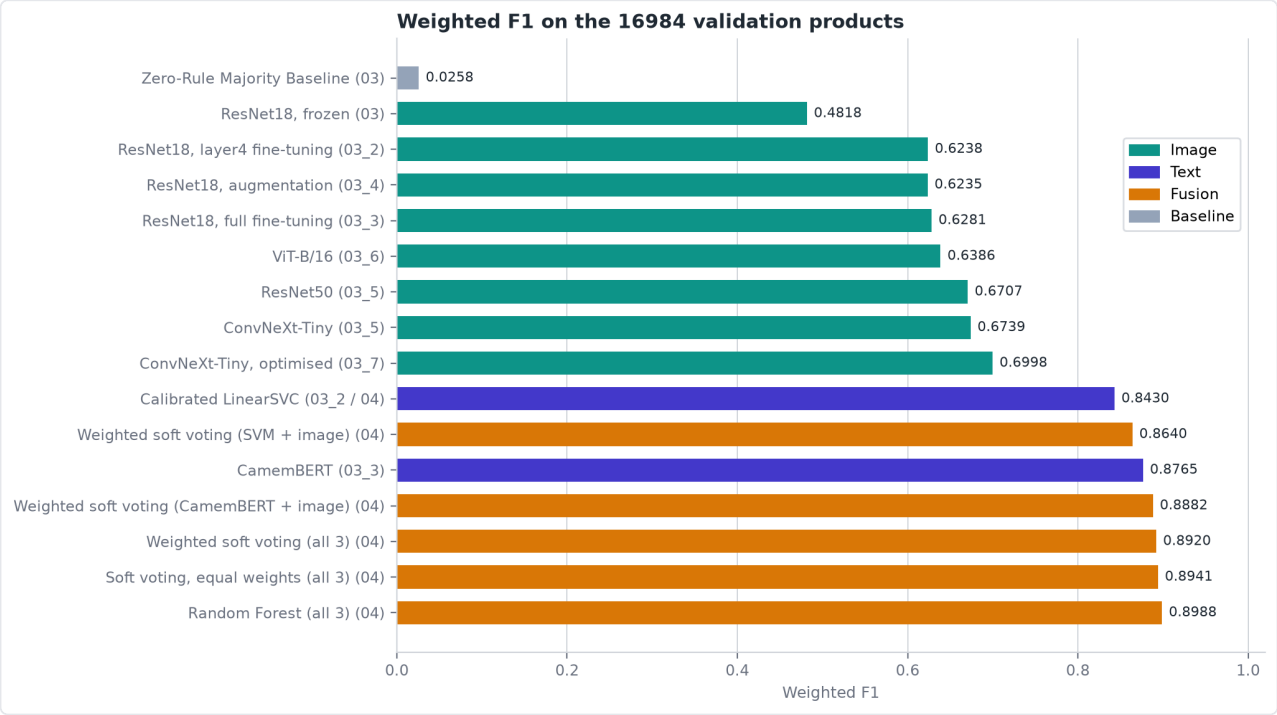


Figure 1: Weighted F1 progression across the majority baseline, image models, text models, and late-fusion methods on the shared 16,984 validation products.

2. Exploratory Data Analysis and Data Audit

2.1 Class Distribution and Imbalance

Rakuten provides **84,916 labelled training products** alongside 13,812 unlabelled challenge test products [notebook `01_dataviz_and_audit.ipynb`]. Across the 27 categories in the labelled set, class sizes are strongly uneven:

- The largest category is **Pool (code 2583)** with **10,209 products** (12.02% of the catalog; 2,042 in validation).
- The smallest category is **Wargames (code 1180)** with **764 products** (0.90% of the catalog; 153 in validation).
- The largest-to-smallest ratio is **13.36 to 1** (13.4:1), and **six categories** contain fewer than 1,000 products (codes `1180`, `1940`, `1301`, `2220`, `60`, and `2905`).
- Because accuracy is dominated by large categories, we report **weighted F1** (the official challenge metric) alongside **macro F1** (unweighted average across all 27 classes).

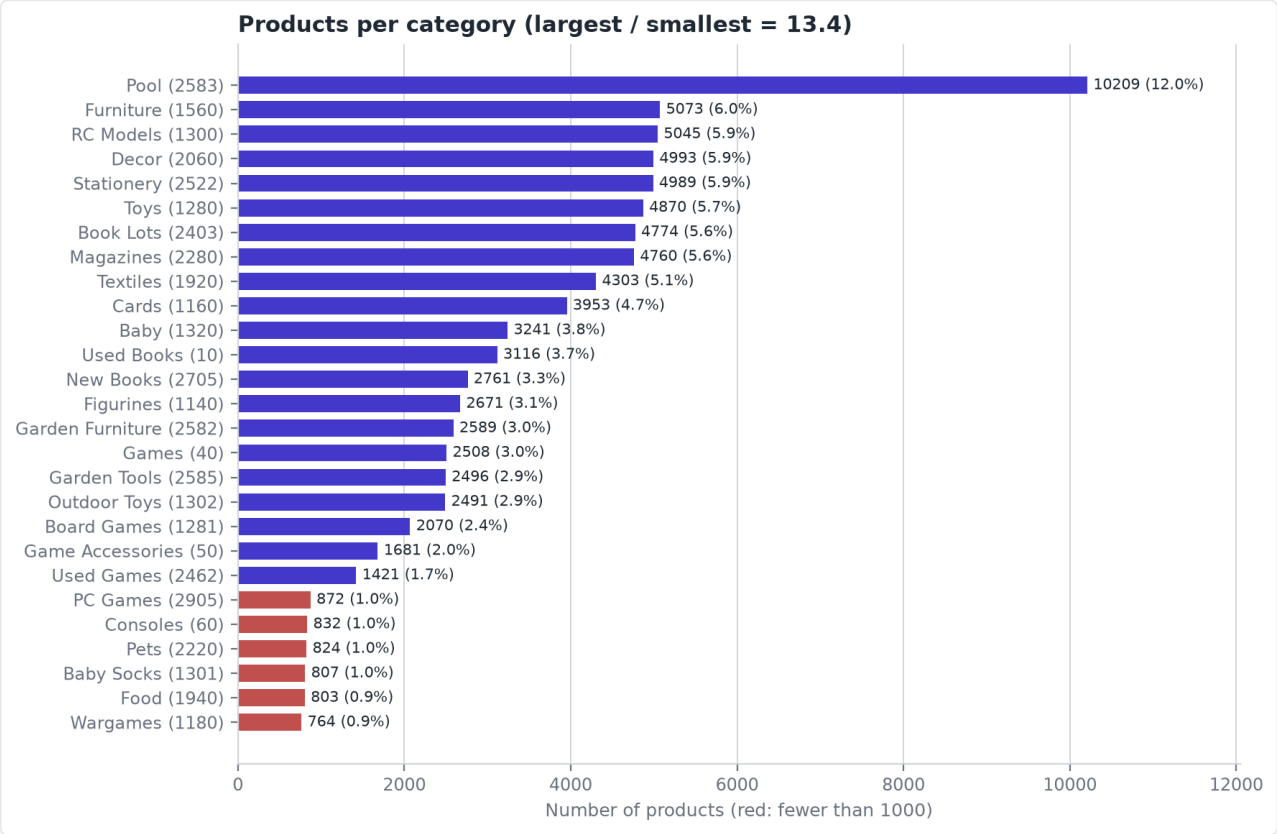


Figure 2: Products per category across the 84,916 labelled listings. Red bars mark the six categories with fewer than 1,000 products.

Note on Category Names: The dataset provides only numeric `prdtypecode` identifiers. All category names in this report and in `src/categories.py` (such as *Used Books*, *Wargames*, or *Pool*) are our own interpretation inferred from inspecting sample titles; they are not official Rakuten labels.

Code	Interpreted Name (`src/categories.py`)	Full Count	Val. Count	Catalog Share	Missing Desc. %
10	Used Books	3,116	623	3.67%	89.2%
40	Games	2,508	502	2.95%	65.5%
50	Game Accessories	1,681	336	1.98%	28.3%
60	Consoles	832	166	0.98%	16.0%
1140	Figurines	2,671	534	3.15%	64.6%
1160	Cards	3,953	791	4.66%	91.2%
1180	Wargames	764	153	0.90%	80.0%
1280	Toys	4,870	974	5.74%	21.9%
1281	Board Games	2,070	414	2.44%	23.8%
1300	RC Models	5,045	1,009	5.94%	23.3%
1301	Baby Socks	807	161	0.95%	13.0%
1302	Outdoor Toys	2,491	498	2.93%	11.1%
1320	Baby	3,241	648	3.82%	34.0%
1560	Furniture	5,073	1,015	5.97%	3.5%
1920	Textiles	4,303	861	5.07%	4.8%
1940	Food	803	161	0.95%	5.6%
2060	Decor	4,993	999	5.88%	5.7%
2220	Pets	824	165	0.97%	8.0%
2280	Magazines	4,760	952	5.61%	93.3%
2403	Book Lots	4,774	955	5.62%	97.4%
2462	Used Games	1,421	284	1.67%	96.3%
2522	Stationery	4,989	998	5.88%	23.8%
2582	Garden Furniture	2,589	518	3.05%	2.4%
2583	**Pool**	**10,209**	**2,042**	**12.02%**	**8.8%**
2585	Garden Tools	2,496	499	2.94%	7.9%
2705	New Books	2,761	552	3.25%	37.1%
2905	PC Games	872	174	1.03%	0.0%

2.2 Text Exploration: Lengths, Missingness, and Noise

Auditing the text columns in notebook `01_dataviz_and_audit.ipynb` established four structural properties:

- **Titles (`designation`):** Present for 100% of products. Word counts range from **3 to 56 words**, with a median of **11 words** and an interquartile range of **7 to 14 words** (mean 11.6). Class medians span from 5 words in New Books (`2705`) to 21 words in Consoles (`60`).
- **Descriptions (`description`):** Missing for **29,800 products (35.09%)** overall and **5,936 products (34.95%)** in validation. Missingness varies sharply by category: PC Games (`2905`) has **0.0%** missing descriptions, whereas Book Lots (`2403`) has **97.4%**, Used Games (`2462`) **96.3%**, Magazines (`2280`) **93.3%**, Cards (`1160`) **91.2%**, and Used Books (`10`) **89.2%**. Where present, descriptions range from 1 to 2,068 words (class medians from 13 words in `1160` to 323 words in

`2905`) and correlate only weakly with title length (Spearman $r = 0.265$). A 128-word cap retains 62.83% of available descriptions in full (67.55% of description words).

- **HTML and Encoding Noise:** Encoded HTML entities (such as `&`, `é`, `'`) appear in **35,030 descriptions (41.25%)**, HTML tags (`<br>`, `<li>`, `<strong>`) in **15,645 (18.42%)**, URLs in 0.24%, and encoding artefacts in 1.08%.
- **Repeated Titles:** After normalizing case, accents, and punctuation, **4,187 products (4.93%)** fall into 1,470 repeated-title groups; **160 groups (497 products)** map to more than one category.

2.3 Image Exploration: Format, White Borders, and Duplicates

- **Coverage and Dimensions:** All **84,916 products (100%)** link to an existing image file. In a random audit sample of 2,000 files, 100% opened cleanly as **500 × 500 pixel RGB JPEGs** [notebook `01_dataviz_and_audit.ipynb`].
- **White Borders:** Many catalog photos place a small product in the center of a wide white canvas. Direct downsampling of the full 500 × 500 image to 224 × 224 shrinks the object itself to a small fraction of the input tensor.
- **Exact File Duplicates:** Computing SHA-256 hashes over all training image files revealed exact duplicate files reused across listings, including generic placeholders shared across conflicting categories [notebook `02_preprocessing_image.ipynb`].

2.4 Exact Input Columns Used by Each Model

The raw dataset contains five columns: `designation`, `description`, `productid`, `imageid`, and target `prdtypecode` (with the original dataframe index preserved as `row_id`). Neither `productid` nor `imageid` is ever passed as a numerical feature; they are used solely to locate the image file on disk.

Model	Modality	Columns Used	Representation
Majority Baseline	BASELINE	`prdtypecode` (train frequencies only)	Predicts constant class `2583` for every validation row.
ResNet18 / ResNet50 / ViT-B/16 / ConvNeXt-Tiny	IMAGE	`imageid`, `productid` (file lookup only)	Loads `image_{imageid}_product_{productid}.jpg` as a 224 × 224 RGB tensor (with `SmartWhiteBorderCrop` in `03_7`).
TF-IDF + LinearSVC (calibrated)	TEXT	`designation`, `description`	Cleaned and joined into `text_combined`; binary word unigram + bigram TF-IDF (460,326 features).
CamemBERT (`camembert-base`)	TEXT	`designation`, `description`	Cleaned and joined into `text_combined`; SentencePiece subword tokens truncated to 128 tokens.
Soft Voting / Random Forest Stacking	FUSION	Out-of-sample class probabilities aligned via `row_id`	27 probabilities per unimodal model (81-dimensional concatenated vector for Random Forest).

3. Preprocessing and Validation Protocol

3.1 Shared 80/20 Split and Training-Only Deduplication

> **Shared Stratified Split:** Notebook `02_preprocessing.ipynb` creates one stratified 80/20 split (**67,932 training products, 16,984 validation products**, `random_state=42`) stored in `data/processed/data_split.csv` and keyed by `row_id`. Every text, image, and fusion notebook evaluates on these exact 16,984 validation rows. The 13,812 unlabelled official test products remain completely separate.

In notebook `02_preprocessing_image.ipynb`, the 16,984 validation rows are kept completely intact so that validation reflects realistic test conditions and aligns 1-to-1 with text models. Exact SHA-256 deduplication is applied **to the training split only**: training rows whose image hash also appears in validation, 676 training rows whose image hash maps to conflicting labels, and duplicate same-label training hashes are removed (**5,620 training rows excluded in total**), leaving **62,312 clean training images** (79,296 total rows in `image_split_clean.csv`).

3.2 How the 16,984 Validation Rows Were Used

To make our evaluation protocol explicit [notebook `04_text_and_image_fusion.ipynb`]:

1. **Unimodal models never train on validation:** TF-IDF + LinearSVC (including its internal 3-fold Platt scaling calibration), CamemBERT, and all image networks were fitted exclusively on the training partition. None of them used the 16,984 validation rows for parameter updates or hyperparameter selection.
2. **Out-of-sample unimodal probabilities:** Each trained unimodal model predicted a 27-class probability vector on each of the 16,984 validation rows. Unimodal validation scores (0.8430 SVM, 0.8765 CamemBERT, 0.6998 ConvNeXt-Tiny) are direct hold-out evaluations on those 16,984 rows.
3. **5-fold cross-validation on the 16,984 validation rows for fusion:** Because we did not carve out a separate meta-training split from the 67,932 training rows (computing out-of-fold CamemBERT probabilities on the training set would have required 5 × 2 hours of transformer training), the fusion step uses **5-fold stratified cross-validation** (`StratifiedKFold(n_splits=5, shuffle=True, random_state=42)`) over the 16,984 validation probability vectors:
   - In each of the 5 folds, soft-voting weights (grid search in steps of 0.1) and the Random Forest meta-classifier (300 trees, `min_samples_leaf=2`) are fitted on **4/5 of the validation set (~13,587 rows)** and predict the held-out **1/5 (~3,397 rows)**.
   - Every validation product's fused prediction is therefore made out-of-fold by a combiner that was not fitted on that product.
   - As a parameter-free check, **equal-weight soft voting** (1/3 each, with zero tuned parameters) reaches **0.8941 weighted F1**, confirming that nearly all of the multimodal gain holds without fitting anything on the validation folds.

3.3 Text Cleaning Pipeline

Notebook `02_preprocessing_text.ipynb` and `src/text_utils.py` apply the same deterministic rules to all text: double `html.unescape` decoding, BeautifulSoup HTML tag removal with space separators, lowercasing, and whitespace normalization while preserving French accents, numbers, units, and model codes. Cleaned `designation` and `description` are joined into `text_combined` (using the title alone when description is missing). On raw text, 18 of 27 categories had HTML tokens (`br`, `li`, `strong`, `39`) among their top six TF-IDF terms; after cleaning, none do. Empirically, LinearSVC scored almost identically on raw vs. cleaned text (0.8276 vs. 0.8274 weighted F1) [notebook `03_text_baseline.ipynb`].

3.4 Image Preprocessing: Smart White-Border Cropping

In notebooks `03_7_image_convnext_optimized.ipynb` and `03_8_image_convnext_gradcam.ipynb`, `SmartWhiteBorderCrop` converts the image to grayscale (`L`) and finds the bounding box of non-white pixels below intensity `WHITE_THRESHOLD = 245`:

- If the object already spans at least 75% of image width or height (`bbox_ratio ≥ 0.75`), the image is left unchanged.

- Otherwise, a square window centered on the bounding box (with `12 px` padding and a minimum crop side of `224 px`) is cut out at native resolution before bicubic resizing to 236, center-cropping to 224 × 224, and ImageNet normalization.

4. Text Classification Models

4.1 Classical TF-IDF Models and Hyperparameter Tuning

Starting from a 50,000-feature TF-IDF + LinearSVC baseline in notebook `03_text_baseline.ipynb` (0.8274 weighted F1; 0.8313 with binary TF-IDF), we tuned the classical pipeline using **3-fold cross-validation on the 67,932 training products only** [notebooks `03_text_baseline.ipynb`, `03_2_text_optimisation.ipynb`]:

- **Removing the feature cap:** The full training vocabulary contains **460,326 word unigrams and bigrams** (`ngram_range=(1, 2)`). In 3-fold training CV, 20k features scored 0.8027, 50k scored 0.8164, 100k scored 0.8231, and keeping all 460,326 features scored **0.8284** (+1.2 points over 50k), retaining rare brand and model identifiers. Only word unigrams and bigrams were evaluated in TF-IDF; character n-grams were not tested.
- **Binary TF-IDF and regularization:** Across 8 grid combinations of `binary` ∈ {`True`, `False`} and $C \in \{0.1, 0.5, 1.0, 2.0\}$, `binary=True` beat standard term counts (0.8284 vs. 0.8235, +0.5 points against a 0.3 CV spread) by preventing repeated seller keywords from dominating, and $C = 1.0$ was optimal.

Holding the chosen TF-IDF representation fixed, three linear classifiers were compared with the same 3-fold cross-validation on the training split [notebook `03_2_text_optimisation.ipynb`]:

Classifier (3-Fold CV on Train Split)	Train CV Weighted F1	Notes
Complement Naive Bayes	0.7600	Fast baseline, 6.8 points below LinearSVC on training CV.
Logistic Regression (SGD)	0.8084	Produces direct probabilities, 2.0 points below LinearSVC on training CV.
LinearSVC ($C = 1.0$)	**0.8284**	**Selected classical text classifier; reaches 0.8436 uncalibrated on validation.**

Evaluated once on the 16,984 validation rows, the tuned uncalibrated LinearSVC reached **0.8436 weighted F1** (0.8330 macro F1) [notebook `03_2_text_optimisation.ipynb`]. Because late fusion requires calibrated probabilities rather than raw margins, notebook `04_text_and_image_fusion.ipynb` wraps LinearSVC in 3-fold sigmoid Platt scaling (`CalibratedClassifierCV(cv=3)`, `min_df=2`) fitted on the training split, yielding **0.8430 weighted F1** (0.8326 macro F1, 2,656 validation errors), matching Figure 1.

4.2 Fine-Tuning CamemBERT

To capture word order and French context beyond bag-of-words n-grams, we fine-tuned **CamemBERT** (`almanach/camembert-base`, 110M parameters) on `text_combined` truncated to 128 tokens [notebook `03_3_text_camembert.ipynb`]:

- **Training configuration:** 2 epochs on the 67,932 training products (~2 hours on an Apple M1 Pro GPU), batch size 16, AdamW (learning rate 5×10^{-5}, weight decay 0.01), 6% linear warmup (509 of 8,492 steps) followed by linear decay, dropout 0.1, and gradient clipping 1.0. Standard BERT fine-tuning hyperparameters were used without tuning on validation.
- **Convergence:** Validation loss decreased from **0.501** after epoch 1 (weighted F1 **0.8527**) to **0.427** after epoch 2, reaching **0.8765 weighted F1** and **0.8594 macro F1** (2,082 errors), confirming no overfitting over two epochs.
- **Comparison with calibrated LinearSVC:** CamemBERT cut validation errors from 2,656 to 2,082 (**574 fewer errors, –21.6%**). Its largest F1 gains occurred on the four book categories: Used Books (+12.3 points), Magazines (+10.9), New Books (+10.7), and Book Lots (+10.0). LinearSVC stayed stronger on Wargames (–8.4 points for CamemBERT) and Outdoor Toys (–2.8), where exact rare product terms (*warhammer*, *heroclix*) decide.

5. Image Models and Grad-CAM Interpretability

5.1 Deep Learning Progression Across Vision Backbones

All image models were trained on the deduplicated 62,312 training images (`image_split_clean.csv`) and evaluated on the complete, unchanged **16,984 validation products** so that every score in Table 5.1 and Figure 1 is directly comparable [notebooks `03_modeling_image.ipynb` and `03_image_fineTuning/03_2` to `03_7`]:

Notebook	Backbone	Training Setup	Weighted F1	Macro F1	Errors (/ 16,984)
`03_modeling_image`	ResNet18	Frozen ImageNet backbone; linear classification head only.	0.4818	0.4231	8,462
`03_2`	ResNet18	Head warmup followed by fine-tuning `layer4` + head.	0.6238	0.5902	6,369
`03_4`	ResNet18	`layer4` + head with ColorJitter, RandomRotation, RandomErasing, label smoothing 0.10.	0.6235	0.5876	6,377
`03_3`	ResNet18	Full-network fine-tuning from `03_2` checkpoint with discriminative LRs.	0.6281	0.5943	6,315
`03_6`	ViT-B/16	Pretrained Vision Transformer; head warmup + final two encoder blocks.	0.6386	0.5982	6,108
`03_5`	ResNet50	Head warmup + final residual block under standard 224 × 224 transforms.	0.6707	0.6415	5,566
`03_5`	ConvNeXt-Tiny	Head warmup + final feature block under standard 224 × 224 transforms.	0.6739	0.6458	5,492
`03_7`	**ConvNeXt-Tiny**	**`SmartWhiteBorderCrop` , full fine-tuning (`features[0..7]` , 4 LR tiers), minority sampler, flip TTA.**	**0.6998**	**0.6752**	**5,043**

- **Partial vs. full ResNet18 fine-tuning:** Unfreezing `layer4` added +14.2 weighted F1 points over the frozen baseline (0.4818 → 0.6238). Unfreezing all ResNet18 layers added only +0.43 points (0.6281) while validation loss rose after epoch 1, indicating early overfitting; heavier augmentation and label smoothing in `03_4` (0.6235) did not improve over `03_2` .
- **Modern CNN vs. ViT:** Under identical partial fine-tuning, ConvNeXt-Tiny (0.6739) outperformed both ResNet50 (0.6707) and ViT-B/16 (0.6386).
- **Optimised ConvNeXt-Tiny (`03_7`) and the role of cropping:** Experiment `03_7` combined `SmartWhiteBorderCrop` , 1 head-warmup epoch followed by 6 full-backbone epochs across `features[0..7]` with four learning-rate tiers (10^{-5}, 3×10^{-5}, 8×10^{-5}, and 10^{-4} under cosine annealing), damped minority handling (`WeightedRandomSampler` with power 0.25 and class loss weights with power 0.15), and horizontal-flip test-time augmentation (TTA). Together, these changes raised ConvNeXt-Tiny by **+2.59 weighted F1 points** (0.6739 → 0.6998) and **+2.94 macro F1 points** (0.6458 → 0.6752). Horizontal-flip TTA alone accounts for **+0.18 weighted F1 points** (0.6980 single-view at epoch 7 → 0.6998 with TTA); the remaining **+2.41 points** come jointly from smart cropping, full-backbone discriminative fine-tuning, and balanced sampling, as cropping was not trained in a separate single-factor ablation.

5.2 Grad-CAM Visual Interpretability Analysis

Notebook `03_8_image_convnext_gradcam.ipynb` inspects the single-view `03_7` ConvNeXt-Tiny checkpoint (0.6980 weighted F1) on the shared validation split using Gradient-weighted Class Activation Mapping (Grad-CAM) attached to the final convolutional stage (`model.features[7]`):

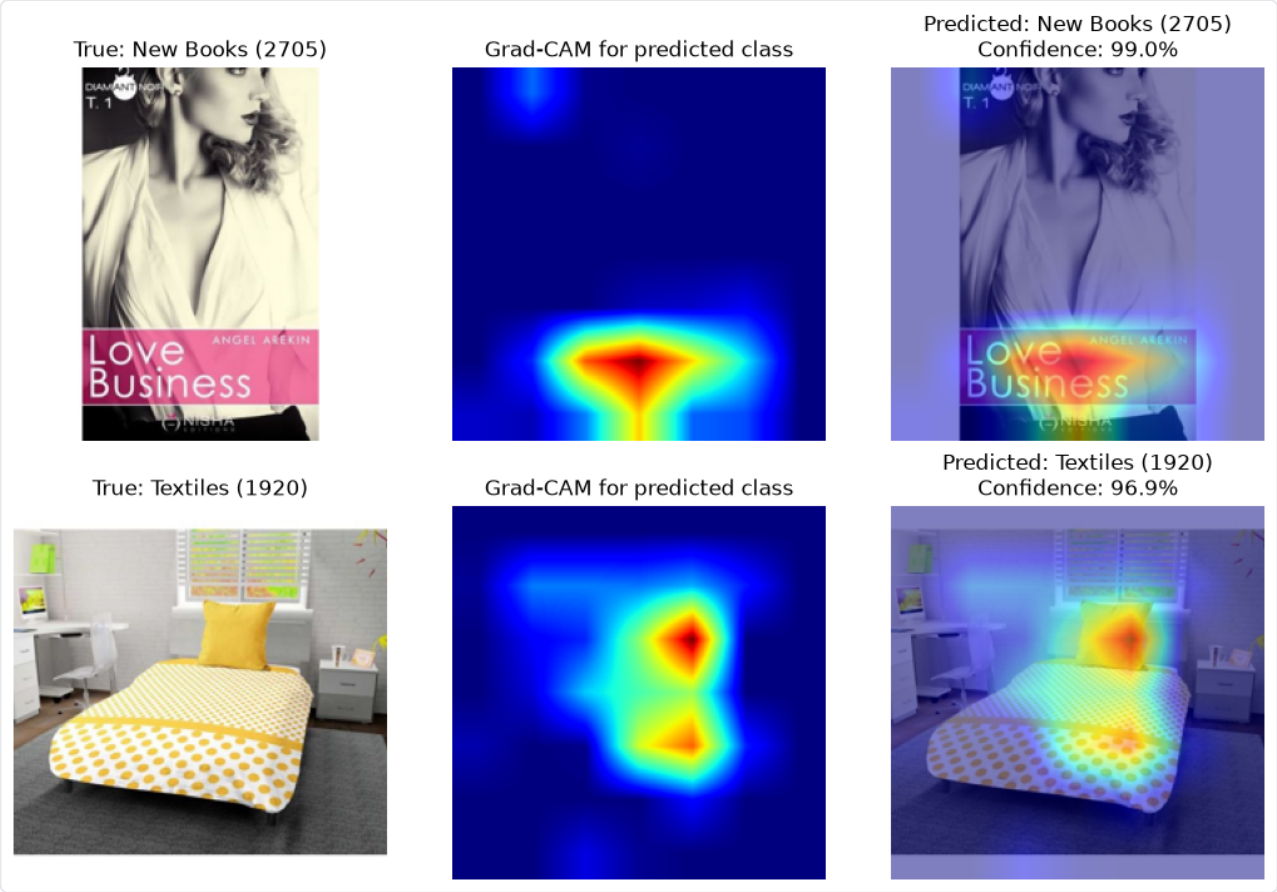


Figure 3: Grad-CAM maps on two correctly classified validation items: New Books (`2705` , 99.0% confidence) and Textiles (`1920` , 96.9% confidence) [notebook `03_8`].

- **Two Correct Predictions (Figure 3):** On `row_id=38611` (New Books `2705` , 99.0% confidence) and `row_id=2315` (Textiles `1920` , 96.9% confidence), positive Grad-CAM attribution focuses directly on the book's title band and on the pillow and patterned duvet rather than surrounding background regions.
- **Controlled Effect of Smart White-Border Cropping (Figure 4):** Keeping resizing, normalization, and model weights fixed on validation item `row_id=21859` (true class Used Books, `10`) isolates `SmartWhiteBorderCrop` at inference time. Without cropping, the book occupies a small central patch, activation spills into the surrounding border, and the model predicts Stationery (`2522` , 73.6% confidence). With smart cropping, the cover is enlarged, Grad-CAM concentrates on the title region, and the prediction switches to Used Books (`10` , 97.2% confidence).
- **Two Representative Misclassifications (Figure 5):** On `row_id=6602` (true class Used Books `10` , predicted New Books `2705` at 94.4% confidence), Grad-CAM highlights the printed book title—confirming the model attends to book cues, even though a clean flat cover image cannot distinguish used from new condition. On `row_id=84339` (a Caran d'Ache crayon tin, true class Toys `1280` , predicted Stationery `2522` at 98.0% confidence), Grad-CAM focuses on the crayons and illustrated crayon tip, reflecting category overlap between creative play sets and art/office supplies rather than background distraction.
- **Interpretability Limitations:** Grad-CAM on `model.features[7]` has coarse spatial resolution, explains single-view predictions rather than the TTA flip average, and illustrates qualitative attention on selected items rather than proving causal reasoning or OCR text reading.

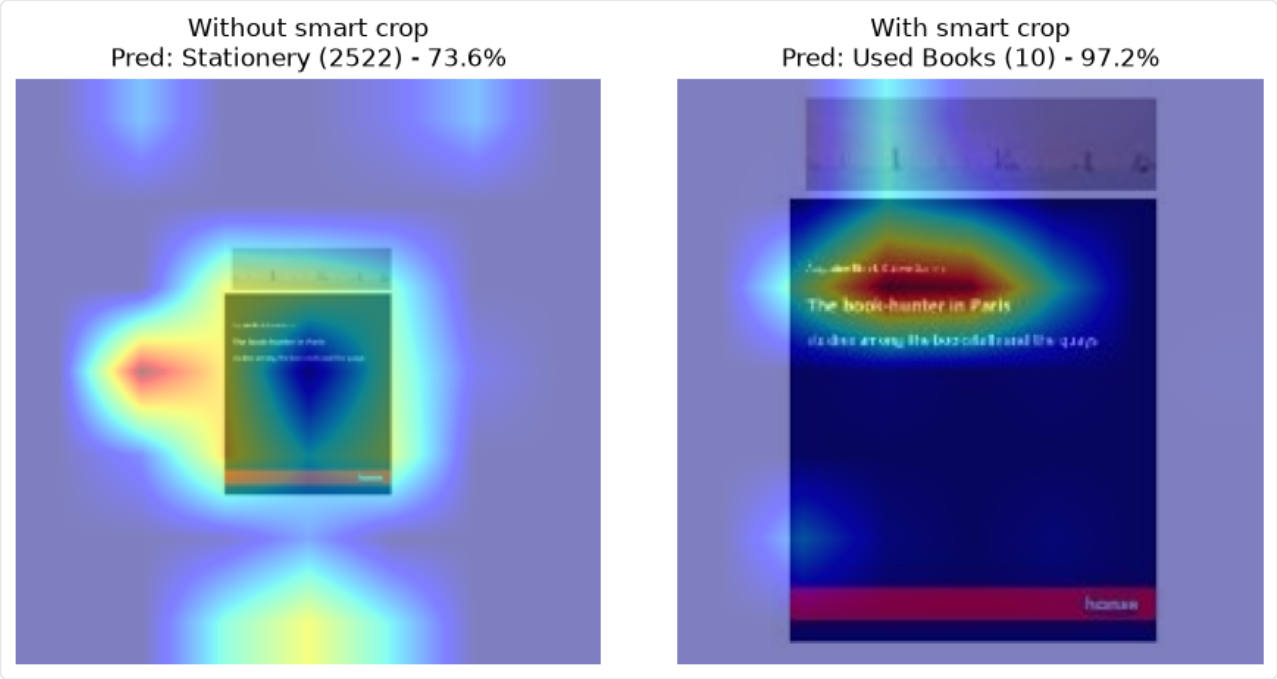


Figure 4: Controlled comparison on validation item `row_id=21859` (Used Books, `10`) [notebook `03_8`]. Without `SmartWhiteBorderCrop`, the model predicts Stationery (`2522`, 73.6%); with cropping, Grad-CAM focuses on the title band and predicts Used Books (`10`, 97.2%).



Figure 5: Grad-CAM on two validation errors [notebook `03_8`]: Used Books (`10`) predicted as New Books (`2705`, 94.4%), and Toys (`1280`) predicted as Stationery (`2522`, 98.0%).

6. Multimodal Fusion and Error Analysis

6.1 Late-Fusion Formulations and Results

Notebook `04_text_and_image_fusion.ipynb` combines the 27-class validation probability vectors from calibrated LinearSVC (P_{SVM}), CamemBERT ($P_{\text{CamemBERT}}$), and optimised ConvNeXt-Tiny (P_{ConvNeXt}), scored out-of-fold with 5-fold stratified cross-validation on the 16,984 validation products:

1. **Soft Voting:** Predicts $\text{argmax}_c \sum_m w_m P_m(c)$, tested both with fixed equal weights (1/3 each) and with weights chosen by grid search (steps of 0.1) on the 4/5 training folds. Across the 5 folds, grid search selected (0.6, 0.4) for SVM + image, (0.5, 0.5) for CamemBERT + image, and (0.4, 0.3, 0.3) in 4 of 5 folds for CamemBERT + SVM + image.
2. **Random Forest Stacking:** Concatenates $[P_{\text{CamemBERT}} \,\|\, P_{\text{SVM}} \,\|\, P_{\text{ConvNeXt}}]$ into an 81-dimensional feature vector and trains a 300-tree Random Forest (`min_samples_leaf=2`) on the 4/5 folds so category-specific trust weights can vary across classes. (In appendix notebook `08_appendix_experiments.ipynb`, Gradient Boosting on SVM + image scored 0.8595 vs. 0.8630 for Random Forest while running ~6× slower, and Random Forest on CamemBERT + image without SVM scored 0.8837.)

Model (`04_fusion_results.csv`)	Modality	Weighted F1	Macro F1	Δ vs. CamemBERT (pts)	Errors (/ 16,984)
ConvNeXt-Tiny (`03_7`)	IMAGE	0.6998	0.6752	–17.67	5,043
Calibrated LinearSVC (`03_2` / `04`)	TEXT	0.8430	0.8326	–3.35	2,656
Weighted Soft Voting (SVM + Image) (`04`)	FUSION	0.8640	0.8548	–1.25	2,278
CamemBERT (`03_3`)	TEXT	0.8765	0.8594	Reference	2,082
Weighted Soft Voting (CamemBERT + Image) (`04`)	FUSION	0.8882	0.8730	+1.17	1,875
Weighted Soft Voting (all 3) (`04`)	FUSION	0.8920	0.8797	+1.54	1,813
Soft Voting, Equal Weights (all 3) (`04`)	FUSION	0.8941	0.8826	+1.76	1,776
Random Forest Stacking (all 3) (`04`)	FUSION	**0.8988**	**0.8856**	**+2.22**	**1,715**

- **Equal vs. tuned soft-voting weights:** Equal weights on all three models (0.8941) slightly beat grid-searched global weights (0.8920), because global scalar weights chosen on 4/5 folds do not improve out-of-fold generalization over 1/3 averaging.
- **Random Forest vs. soft voting:** Random Forest stacking reaches **0.8988 weighted F1** (98 fewer errors than weighted soft voting and 61 fewer than equal-weight soft voting) and outperforms weighted soft voting on 18 of 27 categories, led by New Books (+4.2 points) and Used Books (+3.1).

6.2 Fusion Gain on Products With vs. Without Description

Splitting the 16,984 validation products into those **with a description (11,048 products, 65.05%)** and those **without a description (5,936 products, 34.95%)** quantifies where multimodal fusion adds the most value:

Model	All Val. (16,984)	With Desc. (11,048)	No Desc. (5,936)	No Desc. – With Desc.
ConvNeXt-Tiny (Image)	0.6998	0.6845	0.7351	+0.0506
Calibrated LinearSVC (Text)	0.8430	0.8768	0.7796	–0.0972
CamemBERT (Text)*	0.8785	0.9021	0.8358	–0.0663
Soft Voting, Equal Weights (all 3)*	0.8948	0.9058	0.8746	–0.0312
Random Forest Stacking (all 3)*	**0.9014**	**0.9152**	**0.8756**	**–0.0396**

*Note: Because probability CSVs under `data/processed/` are git-ignored, the subset breakdown in Section 6.2 and confusion matrix in Section 6.4 are computed directly from the saved local CUDA CamemBERT checkpoint (`models/camembert` , trained with the exact `03_3` settings, scoring 0.8785 overall / 2,044 errors vs. 0.8765 / 2,082 errors from the Apple M1 run recorded in `04_fusion_results.csv`).

> **Effect of Missing Descriptions:** Without a description, LinearSVC drops by **9.72 F1 points** (0.8768 → 0.7796) and CamemBERT drops by **6.63 points** (0.9021 → 0.8358). ConvNeXt-Tiny is actually *stronger* on products without descriptions (0.7351 vs. 0.6845) because missing descriptions are concentrated in visually distinct book, magazine, card, and game categories. As a result, Random Forest fusion gains **+3.98 F1 points over CamemBERT** on listings without descriptions (0.8358 → 0.8756), compared to **+1.31 points** when descriptions are present (0.9021 → 0.9152).

6.3 Per-Category Synergy Analysis

Comparing Random Forest fusion against CamemBERT per category [`models/outputs/04_fusion_per_class.csv` , notebook `04_text_and_image_fusion.ipynb`] matches Figure 6 directly:

- At the 0.1-point precision displayed in Figure 6, **24 of the 27 categories improve**, **2 remain at +0.0** (PC Games `2905` at +0.0000 and RC Models `1300` at +0.0005), and **1 decreases slightly** (Baby Socks `1301` , –0.3 points / –0.0033).
- **Largest category gains:** New Books (`2705` , **+10.8 points**, 0.8177 → 0.9258), Used Books (`10` , **+9.9 points**, 0.6889 → 0.7878), Wargames (`1180` , **+9.3 points**, 0.6136 → 0.7063, where LinearSVC supplies exact game keywords), Games (`40` , **+5.9 points**, 0.7953 → 0.8546), and Outdoor Toys (`1302` , **+4.3 points**, 0.8097 → 0.8527).

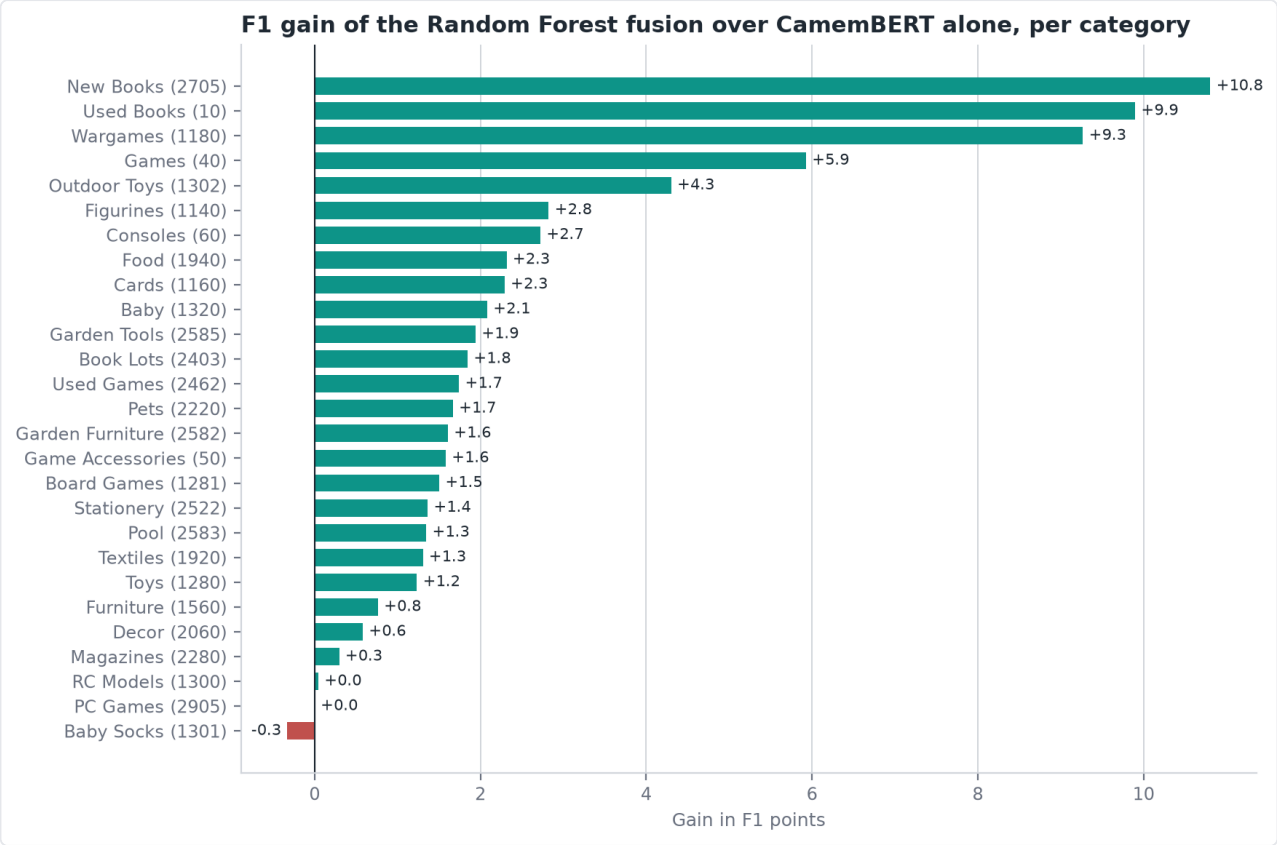


Figure 6: Per-category F1 delta (in F1 points) of Random Forest fusion over CamemBERT alone [`04_fusion_per_class.csv`]. 24 categories show a positive gain (≥ +0.3), 2 round to +0.0 (`2905` , `1300`), and 1 drops slightly (`1301` , −0.3).

6.4 Confusion Matrix and Remaining Error Pairs

Figure 7 shows the row-normalized out-of-fold confusion matrix of the 5-fold Random Forest fusion across all 27 categories on the 16,984 validation products:

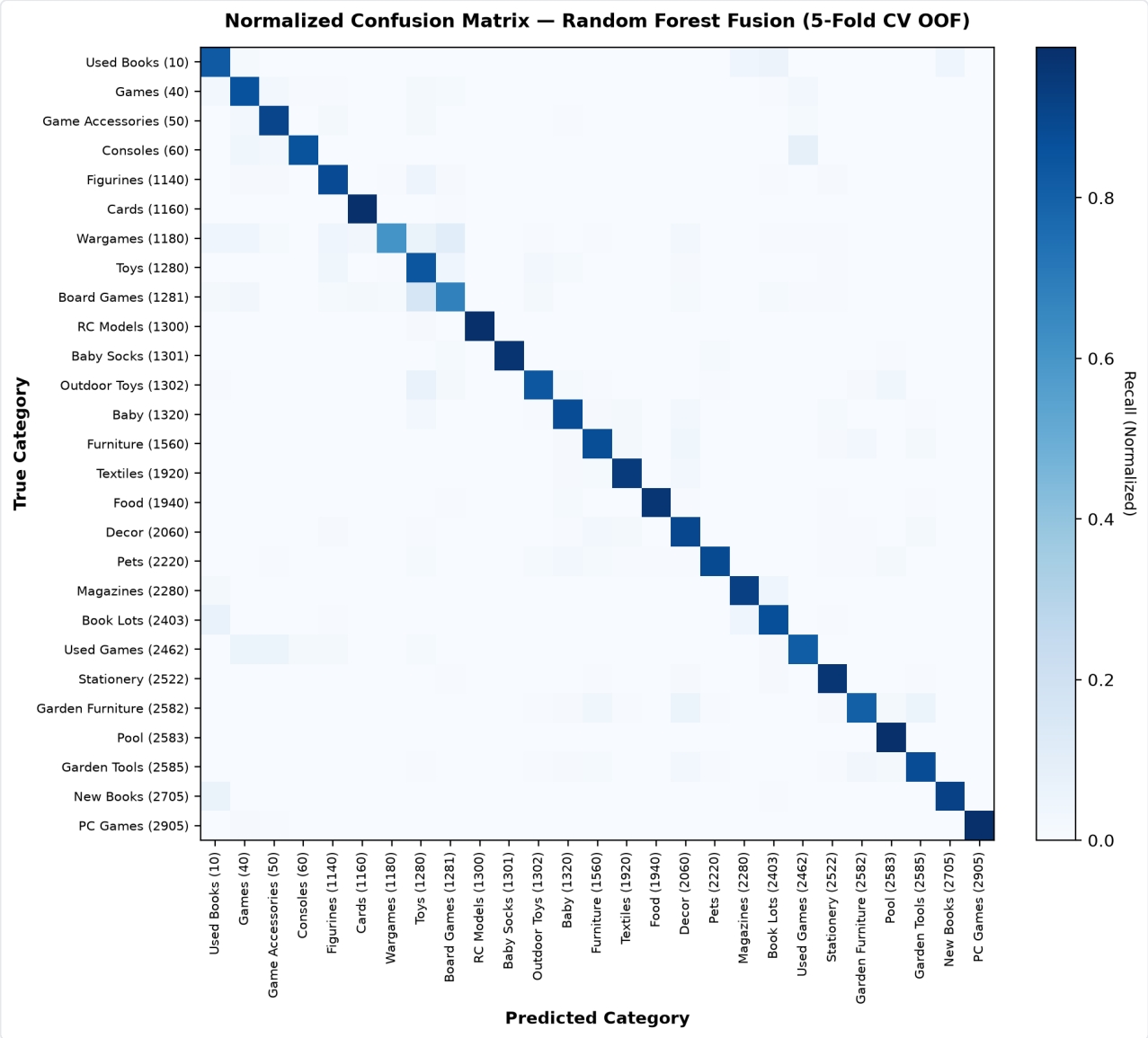


Figure 7: Row-normalized (recall) out-of-fold confusion matrix of the Random Forest fusion across the 27 categories on the 16,984 validation products.

True Category	Predicted Category	OOF Errors	Source of Confusion
Book Lots (`2403`)	Used Books (`10`)	65	Multi-book bundles vs. single used books with missing descriptions.
Board Games (`1281`)	Toys (`1280`)	65	Boxed children's games and toy sets with overlapping brands and packaging.
Toys (`1280`)	Board Games (`1281`)	44	Reciprocal overlap between family play sets and tabletop games.
Magazines (`2280`)	Book Lots (`2403`)	43	Multi-issue magazine lots sharing bundle vocabulary (*lot*, *numéros*).
Furniture (`1560`)	Decor (`2060`)	39	Small storage, shelving, and decorative home furnishings.
Book Lots (`2403`)	Magazines (`2280`)	38	Serialized comics and periodicals sold as sets.
New Books (`2705`)	Used Books (`10`)	35	Same book titles when neither text nor cover photo clearly reveals condition.
Toys (`1280`)	Figurines (`1140`)	35	Licensed character figures straddling children's toys and collectibles.
Used Books (`10`)	Book Lots (`2403`)	35	Single volumes mentioning series/edition terms (*tome*, *collection*).

Remaining errors cluster almost entirely within neighbouring categories of the same product family (books/magazines, toys/games/figurines, and furniture/decor). Unrelated domains (such as Pool `2583`, PC Games `2905`, RC Models `1300`, or Cards `1160`) exceed 0.97–0.99 F1.

7. Text and Multimodal Feature Attribution

7.1 LinearSVC Weights: Product Terms, Seller Shortcuts, and Book Ambiguity

Inspecting the highest LinearSVC weights per category in notebook `05_interpretability.ipynb` shows both why TF-IDF is strong and where it is fragile:

- **Distinctive product terms:** Most categories are anchored by clear domain words: *piscine* and *intex* for Pool (`2583`), *chien* and *laisse* for Pets (`2220`), *warhammer* and *heroclix* for Wargames (`1180`), and *bébé* and *poussette* for Baby (`1320`).
- **Seller names and writing style:** Several high-weight features are seller usernames or store tags (such as *sweetheart93*, *germenly*, *lujuik*, *cdaffaires*), while New Books relies on publisher blurb phrases (*ce livre*, *auteur*). These shortcuts help on existing catalog sellers but may not generalize to new sellers.
- **Shared book vocabulary and short-title failure:** Across the four book classes (Figure 8), generic words like *roman* carry nearly identical weights (0.24 to 0.58), and *tome* / *édition* point to both Used Books and Book Lots. In the case study of validation title *"la promenade au parc"* (true class Used Books `10` , no description), no book keyword appears at all; both class margins are negative (−0.50 for Book Lots vs. −0.68 for Used Books), causing a low-confidence misclassification.

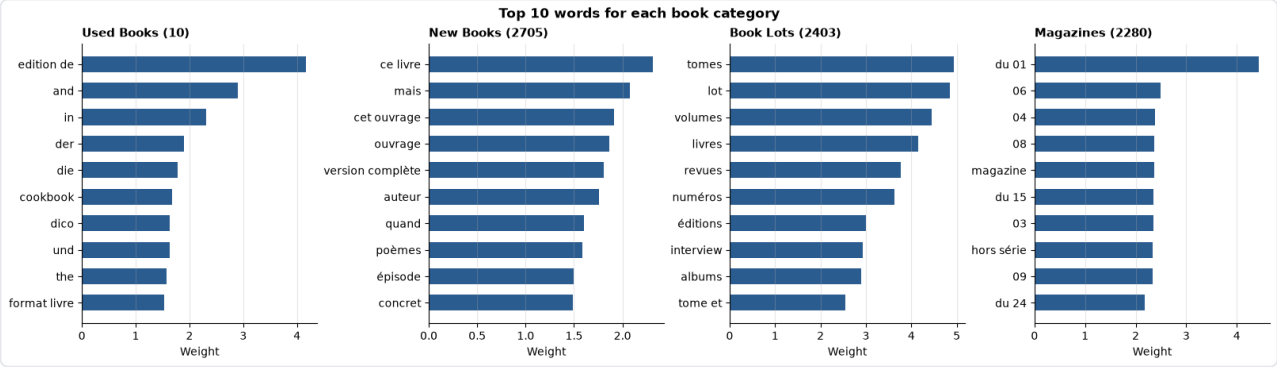


Figure 8: Top LinearSVC feature weights across the four book categories [notebook `05_interpretability.ipynb`]. Specific markers (*lot*, *tomes*, dates) separate subsets, whereas generic literary words are shared.

7.2 Random Forest Meta-Feature Importance and Image Ablation

Notebook `05_interpretability.ipynb` (Section C) evaluates how the Random Forest meta-classifier uses its 81 probability inputs (Figures 9 and 10):

- **Gini importance share (Figure 9):** CamemBERT probabilities account for **66%** of total tree split importance, calibrated LinearSVC accounts for **23%**, and ConvNeXt-Tiny accounts for **11%**.
- **5-fold CV image ablation (Figure 10):** Training the exact same 5-fold Random Forest on text probabilities alone ($[P_{\text{CamemBERT}} \,||\, P_{\text{SVM}}]$) yields **0.8873 weighted F1**, versus **0.8988** when ConvNeXt-Tiny probabilities are included (**+1.15 points**, rounded to +1.2).
- **Where the image stream drives the gain:** Almost all of the image contribution comes from separating **New Books (+11.4 F1 points)** and **Used Books (+10.1 points)**—where clean publisher cover renders vs. seller photos of worn copies distinguish condition—plus **Games (+4.3 points)**. Across most other categories, adding the image changes F1 by less than ±0.7 points.

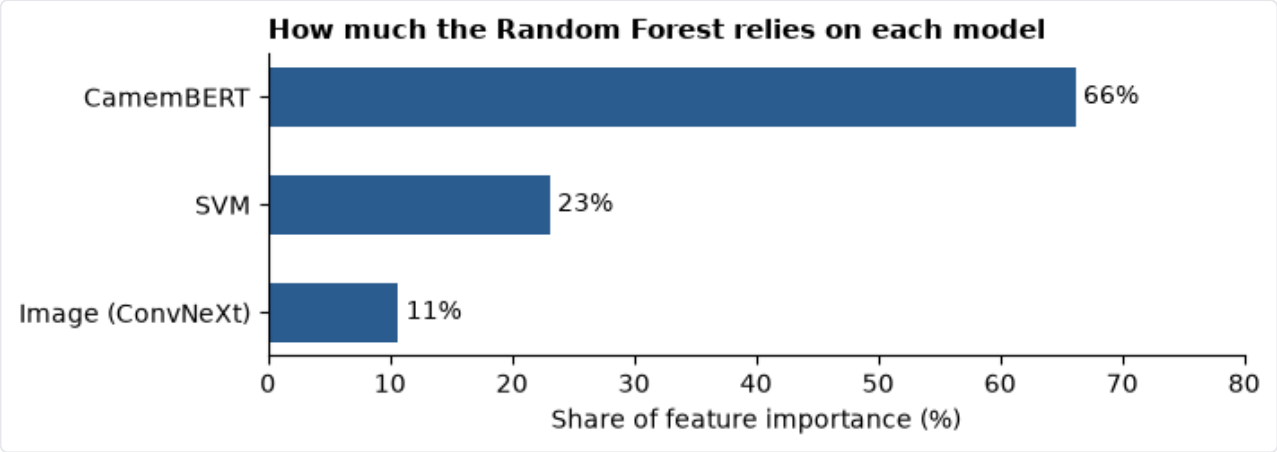


Figure 9: Share of Random Forest Gini feature importance contributed by each unimodal model's 27 probability outputs [notebook `05_interpretability.ipynb`].

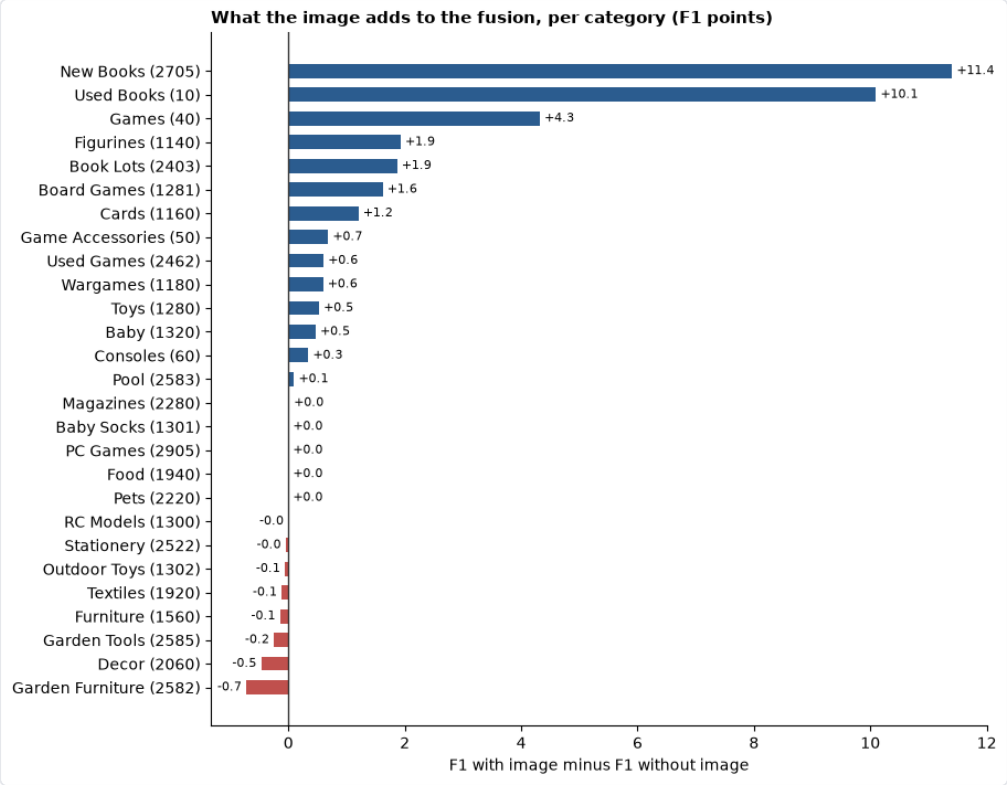


Figure 10: Per-category F1 difference (in F1 points) between the 5-fold Random Forest with the image model and without the image model [notebook `05_interpretability.ipynb`].

8. Practical Recommendations, Limitations, and Conclusion

8.1 Choosing a Pipeline Configuration

Depending on compute budget and maintenance constraints, our experiments support three practical operating points:

Operating Point	Pipeline	Weighted F1	Macro F1	Practical Trade-off
CPU-Only Baseline	Calibrated LinearSVC (TF-IDF)	0.8430	0.8326	Trains in seconds on CPU; requires no GPU for training or inference.
Zero-Parameter Fusion	Equal-Weight Soft Voting (all 3)	0.8941	0.8826	Averages the three probability vectors (1/3 each) with no meta-classifier to fit or maintain.
Best Accuracy	**Random Forest Stacking (all 3)**	**0.8988**	**0.8856**	**Learns category-specific modality trust; cuts 61 more errors than equal-weight voting, mainly on books.**

From a catalog-operations standpoint, two workflow measures follow directly from our error analysis:

- **Prompt sellers for short descriptions:** Absence of a description reduces CamemBERT F1 by 6.63 points and LinearSVC F1 by 9.72 points. Requiring even a brief condition or format note at listing creation addresses the largest source of book and game confusions.
- **Human-in-the-loop confirmation on low-margin predictions:** In `03_text_baseline.ipynb`, 36% of errors had a decision margin below 0.20, and remaining fusion errors concentrate in sibling categories (such as New vs. Used Books or Toys vs. Board Games). Showing sellers the top-2 suggested categories when ensemble probability is split avoids silent miscategorization.

8.2 Limitations

- **Local validation vs. hidden challenge test set:** All scores are measured on our locked 16,984-product validation split; Rakuten's official 13,812-product test set has hidden labels.
- **Fusion fitted via cross-validation on validation rows:** While out-of-fold 5-fold CV prevents any meta-model from predicting rows it saw during fitting, generating out-of-fold unimodal probabilities on the 67,932 training rows would allow fitting the Random Forest without touching validation labels at all.
- **Untuned transformer and unisolated cropping factor:** CamemBERT used standard hyperparameters (2 epochs, 128 tokens) due to its ~2-hour training cost per run, and `03_7` combined `SmartWhiteBorderCrop` with full-backbone fine-tuning and balanced sampling rather than ablating cropping in a separate training run.
- **Seller-specific vocabulary:** Linear text features partially latch onto recurring seller usernames, requiring periodic retraining and monitoring on new marketplace sellers.

8.3 Conclusion

Across 84,916 Rakuten France product listings, systematic progression from exploration and shared-split hygiene (`01` – `02`) through classical and transformer text models (`03_text_baseline` , `03_2` , `03_3`), fine-tuned CNN/ViT backbones with smart white-border cropping and Grad-CAM auditing (`03_image_fineTuning/03_2` – `03_8`), and out-of-fold late fusion (`04` – `05`) raised validation performance from a **0.0258 weighted F1** majority baseline to **0.8988 weighted F1** (**0.8856 macro F1**). Text provides the backbone of the system (CamemBERT 0.8765, LinearSVC 0.8430), LinearSVC preserves exact rare product keywords (such as in Wargames, +9.3 points in fusion), and the optimised ConvNeXt-Tiny image stream (0.6998) supplies the visual condition and packaging cues needed when descriptions are missing (+3.98 points over CamemBERT on no-description listings, and +10.8 / +9.9 points overall on New and Used Books).